

Notes: Bertrand, Duflo & Mullainathan (QJE, 2004)

How Much Should We Trust Differences-in-Differences Estimates?

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1 Big picture

- **Question.** How reliable are conventional standard errors in differences-in-differences designs when the outcome is serially correlated and treatment is persistent?
- **Core problem.** DD treatment variables usually change once and then stay on. Outcomes such as wages and employment are also persistent. Combining persistent regressors with persistent residuals makes naive standard errors too small.
- **Main diagnostic.** The paper randomly creates placebo laws in CPS state-year data. Since the laws are fake, a 5% test should reject about 5% of the time.
- **Magnitude.** In CPS wage data, conventional specifications reject the no-effect null roughly 44–49% of the time in aggregated state-year data, and 67.5% in individual-level wage data.
- **Solution ranking.** Naive parametric AR corrections perform poorly. Block bootstrap works with many groups. Collapsing data into pre/post periods works even with few groups but loses power. Cluster-robust covariance matrices work well with enough clusters.
- **Practical takeaway.** Do not trust conventional DD standard errors in long panels with persistent outcomes. Cluster by treatment unit or use aggregation/randomization-style inference, especially with few clusters.

2 Data and measurement

2.1 Survey of published DD papers

The authors survey 92 DD papers in six major journals between 1990 and 2000. Sixty-nine use more than two periods of data; four collapse to before/after periods; 65 have a potential serial-correlation problem; and only five use an explicit correction. Eighty papers have a potential clustering problem, but only 36 address it.

Interpretation: The survey establishes that the problem is common in applied DD work.

2.2 CPS data and placebo laws

The main exercise uses women aged 25–50 in the fourth interview month of the CPS Merged Outgoing Rotation Group from 1979 to 1999. The main outcome is log weekly earnings. The authors randomly choose half the states to be treated and randomly draw a law year between 1985 and 1995.

Interpretation: Because the law is fake, rejection under the null measures whether the inference procedure is correctly sized.

2.3 Correction methods

The paper evaluates parametric AR corrections, block bootstrap, simple aggregation, residual aggregation, empirical variance-covariance estimation, and cluster-robust covariance estimation.

Interpretation: A method must control false positives without destroying too much power.

3 Models and empirical specifications

3.1 Canonical DD regression

$$Y_{ist} = A_s + B_t + cX_{ist} + \beta I_{st} + \varepsilon_{ist}.$$

Interpretation: The coefficient β is the DD treatment effect. State and year fixed effects absorb permanent state differences and common shocks.

3.2 Why serial correlation creates false significance

$$\text{Var}(\hat{\beta}) \text{ depends on } \text{Cov}(I_{st}\varepsilon_{st}, I_{s\tau}\varepsilon_{s\tau}).$$

Interpretation: If both treatment and residuals are persistent, the effective number of independent observations is much smaller than the number of state-year cells.

3.3 Placebo-law simulation

For each simulation: assign treated states, draw law year, estimate DD, record $1\{|t| > 1.96\}$.

Interpretation: With fake laws, a correctly sized 5% test should reject roughly 5% of the time.

3.4 Cluster-robust covariance matrix

$$\hat{V}_{cluster} = (X'X)^{-1} \left(\sum_s X'_s \hat{u}_s \hat{u}'_s X_s \right) (X'X)^{-1}.$$

Interpretation: Clustering treats the state as the independent sampling unit and allows arbitrary serial correlation within states.

3.5 Aggregation method

Outcomes or residuals are collapsed into two observations per treated unit: pre-treatment and post-treatment.

Interpretation: Collapsing removes the long time-series dimension that creates misleading precision, but it reduces power.

4 Identification and empirical design

The fake laws have no causal effect, so statistically significant placebo effects reveal inference failure rather than treatment effects. CPS data provide a realistic panel with persistent outcomes, while AR(1) simulations show that overrejection is not merely a CPS artifact. The paper is about inference, not about whether DD point estimates are unbiased.

5 Tables and figures

5.1 Table I: survey of DD papers

Read: 92 DD papers are surveyed; 65 have a potential serial-correlation problem; only five correct for it; the average time span is 16.5 periods.

Interpretation: This table motivates the whole paper: the ingredients for misleading DD inference were common in published work.

Economic message: Serial correlation was a practical problem in applied research, not an obscure technical concern.

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Number of DD papers	92
Number with more than 2 periods of data	69
Number which collapse data into before-after	4
Number with potential serial correlation problem	65
Number with some serial correlation correction	5
GLS	4
Arbitrary variance-covariance matrix	1
Distribution of time span for papers with more than 2 periods	Average 16.5 Percentile Value
	1% 3
	5% 3
	10% 4
	25% 5.75
	50% 11
	75% 21.5
	90% 36
	95% 51
	99% 83
Most commonly used dependent variables	Number
Employment	18
Wages	13
Health/medical expenditure	8
Unemployment	6
Fertility/teen motherhood	4
Insurance	4
Poverty	3
Consumption/savings	3
Informal techniques used to assess endogeneity	Number
Graph dynamics of effect	15
See if effect is persistent	2
DDD	11
Include time trend specific to treated states	7
Look for effect prior to intervention	3
Include lagged dependent variable	3
Number with potential clustering problem	80
Number which deal with it	36

^a Data come from a survey of all articles in six journals between 1990 and 2000: the *American Economic Review*, the *Industrial Labor Relations Review*, the *Journal of Labor Economics*, the *Journal of Political Economy*, the *Journal of Public Economics*, and the *Quarterly Journal of Economics*. We define an article as "Difference-in-Difference" if it (1) examines the effect of a specific intervention and (2) uses units unaffected by the intervention as a control group.

number of periods used is 16.5, and the median is 11. More than 75 percent of the papers use more than five periods of data.⁷

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A. CPS DATA				
Data	$\hat{\rho}_1, \hat{\rho}_2, \hat{\rho}_3$	Modifications	Rejection rate	
			No effect	2% effect
1) CPS micro, log wage			.675 (.027)	.855 (.020)
2) CPS micro, log wage		Cluster at state-year level	.44 (.029)	.74 (.025)
3) CPS agg, log wage	.509, .440, .332		.435 (.029)	.72 (.026)
4) CPS agg, log wage	.509, .440, .332	Sampling w/replacement	.49 (.025)	.663 (.024)
5) CPS agg, log wage	.509, .440, .332	Serially uncorrelated laws	.05 (.011)	.988 (.006)
6) CPS agg, employment	.470, .418, .367		.46 (.025)	.88 (.016)
7) CPS agg, hours worked	.151, .114, .063		.265 (.022)	.280 (.022)
8) CPS agg, changes in log wage	-.046, .032, .002		0 (.007)	.978 (.007)
B. MONTE CARLO SIMULATIONS WITH SAMPLING FROM AR(1) DISTRIBUTION				
Data	ρ	Modifications	Rejection rate	
			No effect	2% effect
9) AR(1)	.8		.373 (.028)	.725 (.026)
10) AR(1)	0		.053 (.013)	.783 (.024)
11) AR(1)	.2		.123 (.019)	.738 (.025)
12) AR(1)	.4		.19 (.023)	.713 (.026)
13) AR(1)	.6		.333 (.027)	.700 (.026)
14) AR(1)	-.4		.008 (.005)	.7 (.026)

^a Unless mentioned otherwise under "Modifications," reported in the last two columns are the OLS rejection rates of the null hypothesis of no effect (at the 5 percent significance level) on the intervention variable for randomly generated placebo interventions as described in text. The data used in the last column were altered to simulate a true 2 percent effect of the intervention. The number of simulations for each cell is at least 200 and typically 400.

^b CPS data are data for women between 25 and 50 in the fourth interview month of the Merged Outgoing Rotation Group for the years 1979 to 1999. In rows 3 to 8 of Panel A, data are aggregated to state-year level cells after controlling for demographic variables (four education dummies and a quartic in age). For each simulation in rows 1 through 3, we use the observed CPS data. For each simulation in rows 4 through 8, the data generating process is the state-level empirical distribution of the CPS data that puts a probability of 1/50 on the different states' outcomes (see text for details). For each simulation in Panel B, the data generating process is an AR(1) model with normal disturbances chosen to match the CPS state female wage variances (see text for details). $\hat{\rho}_i$ refer to the estimated autocorrelation parameter of lag i ; ρ refers to the autocorrelation parameter in the AR(1) model.

5.2 Table II: placebo laws and overrejection

Read: Conventional DD rejects fake laws at extremely high rates: 67.5% in individual CPS wages and roughly 44–49% in state-year wage panels. Making laws serially uncorrelated brings rejection down to 5%.

Interpretation: The table isolates the mechanism: persistent outcomes plus persistent treatment variables create false precision.

Economic message: Naive DD can turn random placebo laws into significant results.

5.3 Table III: varying number of groups and periods

Read: More time periods worsen overrejection. With 50 states and 21 years, CPS rejection is 49%; with 3 periods it is about 4.8%.

Interpretation: Long panels create the illusion of precision when serial correlation is ignored.

Economic message: More observations can make inference worse if dependence is not modeled.

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TABLE III
VARYING N AND T

Data	N	T	Rejection rate	
			No effect	2% effect
A. CPS DATA				
1) CPS aggregate	50	21	.49 (.025)	.663 (.024)
2) CPS aggregate	20	21	.39 (.024)	.54 (.025)
3) CPS aggregate	10	21	.443 (.025)	.510 (.025)
4) CPS aggregate	6	21	.383 (.025)	.433 (.025)
5) CPS aggregate	50	11	.20 (.020)	.638 (.024)
6) CPS aggregate	50	7	.15 (.017)	.635 (.024)
7) CPS aggregate	50	5	.078 (.013)	.5 (.025)
8) CPS aggregate	50	3	.048 (.011)	.363 (.024)
9) CPS aggregate	50	2	.055 (.011)	.28 (.022)
B. MONTE CARLO SIMULATIONS WITH SAMPLING FROM AR(1) DISTRIBUTION				
10) AR(1), $\rho = .8$	50	21	.35 (.028)	.638 (.028)
11) AR(1), $\rho = .8$	20	21	.35 (.028)	.538 (.029)
12) AR(1), $\rho = .8$	10	21	.3975 (.028)	.505 (.029)
13) AR(1), $\rho = .8$	6	21	.393 (.028)	.5 (.029)
14) AR(1), $\rho = .8$	50	11	.335 (.027)	.588 (.028)
15) AR(1), $\rho = .8$	50	5	.175 (.022)	.5525 (.029)
16) AR(1), $\rho = .8$	50	3	.09 (.017)	.435 (.029)
17) AR(1), $\rho = .8$	50	50	.4975 (.029)	.855 (.020)

a. Reported in the last two columns are the OLS rejection rates of the null hypothesis of no effect (at the 5 percent significance level) on the intervention variable for randomly generated placebo interventions as described in text. The data used in the last column were altered to simulate a true 2 percent effect of the intervention. The number of simulations for each cell is typically 400 and at least 200.

b. CPS data are data for women between 25 and 50 in the fourth interview month of the Merged Outgoing Rotation Group for the years 1979 to 1999. The dependent variable is log weekly earnings. Data are aggregated to state-year level cells after controlling for the demographic variables (four education dummies and a quartic in age). For each simulation in Panel A, the data generating process is the state-level empirical distribution of the CPS data that puts a probability of 1/50 on the different states' outcomes (see text for details). For each simulation in Panel B, the data generating process is an AR(1) model with normal disturbances chosen to match the CPS state female wage variances (see text for details). ρ refers to the autocorrelation parameter in the AR(1) data generating process.

c. All regressions also include, in addition to the intervention variable, state and year fixed effects.

d. Standard errors are in parentheses and are computed using the number of simulations.

e. N refers to the number of states used in the simulation and T refers to the number of years.

Data	Technique	Estimated $\hat{\rho}_1$	Rejection rate	
			No effect	2% Effect
A. CPS DATA				
1) CPS aggregate	OLS		.49 (.025)	.663 (.024)
2) CPS aggregate	Standard AR(1) correction	.381	.24 (.021)	.66 (.024)
3) CPS aggregate	AR(1) correction imposing $\rho = .8$		.18 (.019)	.363 (.024)
B. OTHER DATA GENERATING PROCESSES				
4) AR(1), $\rho = .8$	OLS		.373 (.028)	.765 (.024)
5) AR(1), $\rho = .8$	Standard AR(1) correction	.622	.205 (.023)	.715 (.026)
6) AR(1), $\rho = .8$	AR(1) correction imposing $\rho = .8$		.06 (.023)	.323 (.027)
7) AR(2), $\rho_1 = .55$ $\rho_2 = .35$	Standard AR(1) correction	.444	.305 (.027)	.625 (.028)
8) AR(1) + white noise, $\rho = .95$, noise/signal = .13	Standard AR(1) correction	.301	.385 (.028)	.4 (.028)

a. Reported in the last two columns are the rejection rates of the null hypothesis of no effect (at the 5 percent significance level) on the intervention variable for randomly generated placebo interventions as described in text. The data used in the last column were altered to simulate a true 2 percent effect of the intervention. The number of simulations for each cell is typically 400 and at least 200.

b. CPS data are data for women between 25 and 50 in the fourth interview month of the Merged Outgoing Rotation Group for the years 1979 to 1999. The dependent variable is log weekly earnings. Data are aggregated to state-year level cells, after controlling for the demographic variables (four education dummies and a quartic in age). For each simulation in Panel A, the data generating process is the state-level empirical distribution of the CPS data that puts a probability of 1/50 on the different states' outcomes (see text for details). For each simulation in Panel B, the distributions from which the data are drawn are chosen to match the CPS state female wage variances (see text for details). "AR(1) + white noise" is the sum of an AR(1) plus an i.i.d. process, where the autocorrelation for the AR(1) component is given by ρ and the relative variance of the components is given by the noise to signal ratio.

c. All regressions include, in addition to the intervention variable, state and year fixed effects.

5.4 Table IV: parametric solutions

Read: Standard AR(1) corrections reduce but do not eliminate overrejection, and imposing high autocorrelation can reduce power.

Interpretation: Parametric corrections depend on knowing the right serial-correlation process and estimating it accurately in short panels.

Economic message: AR(1)-style fixes are fragile in realistic DD settings.

5.5 Table V: block bootstrap

Read: Block bootstrap works well with 50 states but deteriorates as the number of states falls.

Interpretation: Resampling whole state histories preserves serial correlation, but the bootstrap needs enough clusters.

Economic message: Bootstrap is useful with many groups, not a cure for few-cluster inference.

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TABLE V
BLOCK BOOTSTRAP

Data	Technique	N	Rejection rate	
			No effect	2% effect
A. CPS DATA				
1) CPS aggregate	OLS	50	.43 (.025)	.735 (.022)
2) CPS aggregate	Block bootstrap	50	.065 (.013)	.26 (.022)
3) CPS aggregate	OLS	20	.385 (.022)	.595 (.025)
4) CPS aggregate	Block bootstrap	20	.13 (.017)	.19 (.020)
5) CPS aggregate	OLS	10	.385 (.024)	.48 (.024)
6) CPS aggregate	Block bootstrap	10	.225 (.021)	.25 (.022)
7) CPS aggregate	OLS	6	.48 (.025)	.435 (.025)
8) CPS aggregate	Block bootstrap	6	.435 (.022)	.375 (.025)
B. AR(1) DISTRIBUTION				
9) AR(1), $\rho = .8$	OLS	50	.44 (.035)	.70 (.032)
10) AR(1), $\rho = .8$	Block bootstrap	50	.05 (.015)	.25 (.031)

a. Reported in the last two columns are the rejection rates of the null hypothesis of no effect (at the 5 percent significance level) on the intervention variable for randomly generated placebo interventions as described in text. The data used in the last column were altered to simulate a true 2 percent effect of the intervention. The number of simulations for each cell is typically 400 and at least 200. The bootstraps involve 400 repetitions.

b. CPS data are data for women between 25 and 50 in the fourth interview month of the Merged Outgoing Rotation Group for the years 1979 to 1999. The dependent variable is log weekly earnings. Data are aggregated to state-year level cells after controlling for the demographic variables (four education dummies and a quartic in age). For each simulation we draw each state's vector from these data with replacement. See text for details. The AR(1) distribution is chosen to match the CPS state female wage variances (see text for details).

c. All CPS regressions also include, in addition to the intervention variable, state and year fixed effects.

d. Standard errors are in parentheses and are computed using the number of simulations.

model. When there is a 2 percent effect, the null of no effect is rejected in 26 percent of the cases in the CPS data and in 25 percent of the cases in the AR(1) data. However, the method performs less well when the number of states declines. The rejection rate is 13 percent with twenty states and 23 percent with ten states. The power of this test also declines quite fast. With

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TABLE VI
IGNORING TIME SERIES DATA

Data	Technique	N	Rejection rate	
			No effect	2% effect
A. CPS DATA				
1) CPS agg	OLS	50	.49 (.025)	.663 (.024)
2) CPS agg	Simple aggregation	50	.053 (.011)	.163 (.018)
3) CPS agg	Residual aggregation	50	.058 (.011)	.173 (.019)
4) CPS agg, staggered laws	Residual aggregation	50	.048 (.011)	.363 (.024)
5) CPS agg	OLS	20	.39 (.025)	.54 (.025)
6) CPS agg	Simple aggregation	20	.050 (.011)	.088 (.014)
7) CPS agg	Residual aggregation	20	.06 (.011)	.183 (.019)
8) CPS agg, staggered laws	Residual aggregation	20	.048 (.011)	.130 (.017)
9) CPS agg	OLS	10	.443 (.025)	.51 (.025)
10) CPS agg	Simple aggregation	10	.053 (.011)	.065 (.012)
11) CPS agg	Residual aggregation	10	.093 (.014)	.178 (.019)
12) CPS agg, staggered laws	Residual aggregation	10	.088 (.014)	.128 (.017)
13) CPS agg	OLS	6	.383 (.024)	.433 (.024)
14) CPS agg	Simple aggregation	6	.068 (.013)	.07 (.013)
15) CPS agg	Residual aggregation	6	.11 (.016)	.123 (.016)
16) CPS agg, staggered laws	Residual aggregation	6	.09 (.014)	.138 (.017)
B. AR(1) DISTRIBUTION				
17) AR(1), $\rho = .8$	Simple aggregation	50	.050 (.013)	.243 (.025)
18) AR(1), $\rho = .8$	Residual aggregation	50	.045 (.012)	.235 (.024)
19) AR(1), $\rho = .8$, staggered laws	Residual aggregation	50	.075 (.015)	.355 (.028)

a. Reported in the last two columns are the rejection rates of the null hypothesis of no effect (at the 5 percent significance level) on the intervention variable for randomly generated placebo interventions as described in text. The data used in the last column were altered to simulate a true 2 percent effect of the intervention. The number of simulations for each cell is typically 400 and at least 200.

b. CPS data are data for women between 25 and 50 in the fourth interview month of the Merged Outgoing Rotation Group for the years 1979 to 1999. The dependent variable is log weekly earnings. Data are aggregated to state-year level cells after controlling for demographic variables (four education dummies and a quartic in age). For each simulation we draw each state's vector from these data with replacement. See text for details. The AR(1)

5.6 Table VI: ignoring time-series data

Read: Simple aggregation and residual aggregation restore size even with fewer states, although power falls.

Interpretation: Aggregation analyzes the effective source of variation: pre/post differences at the treated-unit level.

Economic message: Conservative inference can be achieved by collapsing the time dimension.

5.7 Table VII: empirical variance-covariance matrix

Read: Empirical covariance estimation works well for 50 states but overrejects as N falls.

Interpretation: It estimates a common within-state covariance structure using cross-state variation, so it needs many groups and homogeneity.

Economic message: Flexible covariance estimation helps in large panels but is unreliable in small-N settings.

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TABLE VII
EMPIRICAL VARIANCE-COVARIANCE MATRIX

Data	Technique	N	Rejection rate	
			No effect	2% effect
A. CPS DATA				
1) CPS aggregate	OLS	50	.49 (.025)	.663 (.024)
2) CPS aggregate	Empirical variance	50	.055 (.011)	.243 (.021)
3) CPS aggregate	OLS	20	.39 (.024)	.54 (.025)
4) CPS aggregate	Empirical variance	20	.08 (.013)	.138 (.017)
5) CPS aggregate	OLS	10	.443 (.025)	.510 (.025)
6) CPS aggregate	Empirical variance	10	.105 (.015)	.145 (.018)
7) CPS aggregate	OLS	6	.383 (.025)	.433 (.025)
8) CPS aggregate	Empirical variance	6	.153 (.018)	.185 (.019)
B. AR(1) DISTRIBUTION				
9) AR(1), $\rho = .8$	Empirical variance	50	.07 (.017)	.25 (.030)

a. Reported in the last two columns are the rejection rates of the null hypothesis of no effect (at the 5 percent significance level) on the intervention variable for randomly generated placebo interventions as described in text. The data used in the last column were altered to simulate a true 2 percent effect of the intervention. The number of simulations for each cell is typically 400 and at least 200.

b. CPS data are data for women between 25 and 50 in the fourth interview month of the Merged Outgoing Rotation Group for the years 1979 to 1999. The dependent variable is log weekly earnings. Data are aggregated to state-year level cells after controlling for demographic variables (four education dummies and a quartic age). For each simulation we draw each state's vector from these data with replacement. See text for details. The AR(1) distribution is chosen to match the CPS state female wage variances (see text for details).

c. All regressions include, in addition to the intervention variable, state and year fixed effects.

d. Standard errors are in parentheses and are computed using the number of simulations.

IV.E. Arbitrary Variance-Covariance Matrix

One obvious limitation of the empirical variance-covariance matrix method discussed above is that it is only consistent under the assumption of cross-sectional homoskedasticity, an assumption that is likely to be violated in practice for many data sets. However, this method can be generalized to an estimator of the

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TABLE VIII
ARBITRARY VARIANCE-COVARIANCE MATRIX

Data	Technique	N	Rejection rate	
			No effect	2% effect
A. CPS DATA				
1) CPS aggregate	OLS	50	.49 (.025)	.663 (.024)
2) CPS aggregate	Cluster	50	.063 (.012)	.268 (.022)
3) CPS aggregate	OLS	20	.385 (.024)	.535 (.025)
4) CPS aggregate	Cluster	20	.058 (.011)	.13 (.017)
5) CPS aggregate	OLS	10	.443 (.025)	.51 (.025)
6) CPS aggregate	Cluster	10	.08 (.014)	.12 (.016)
7) CPS aggregate	OLS	6	.383 (.024)	.433 (.025)
8) CPS aggregate	Cluster	6	.115 (.016)	.118 (.016)
B. AR(1) DISTRIBUTION				
9) AR(1), $\rho = .8$	Cluster	50	.045 (.012)	.275 (.026)
10) AR(1), $\rho = 0$	Cluster	50	.035 (.011)	.74 (.025)

a. Reported in the last two columns are the rejection rates of the null hypothesis of no effect (at the 5 percent significance level) on the intervention variable for randomly generated placebo interventions as described in text. The data used in the last column were altered to simulate a true 2 percent effect of the intervention. The number of simulations for each cell is typically 400 and at least 200.

b. CPS data are data for women between 25 and 50 in the fourth interview month of the Merged Outgoing Rotation Group for the years 1979 to 1999. The dependent variable is log weekly earnings. Data are aggregated to state-year level cells after controlling for demographic variables (four education dummies and a quartic in age). For each simulation we draw each state's vector from these data with replacement. See text for details. The AR(1) distribution is chosen to match the CPS state female wage variances (see text for details).

c. All regressions include, in addition to the intervention variable, state and year fixed effects.

d. Standard errors are in parentheses and are computed using the number of simulations.

the empirical variance-covariance matrix correction method, less extreme than with block bootstraps, but higher than with the time

5.8 Table VIII: arbitrary variance-covariance matrix / clustering

Read: Cluster-robust inference reduces CPS no-effect rejection to 6.3% with 50 states and performs well in AR(1) data.

Interpretation: Clustering by state directly allows arbitrary within-state correlation. The main weakness is small-cluster bias.

Economic message: Cluster at the treatment-unit level when enough clusters exist; use small-sample alternatives when clusters are few.

6 Short summary

- DD is vulnerable to serial correlation because treatments and outcomes are often persistent.
- Placebo laws in CPS wage data show severe overrejection under conventional standard errors.
- Parametric corrections are fragile.
- Block bootstrap and cluster-robust covariance work with many groups.
- Aggregation works even with few groups but has lower power.
- The central lesson is to base inference on the level and persistence of the identifying variation.

7 Conclusion

7.1 Core conclusion

Conventional DD standard errors can be severely misleading in long panels with persistent treatment and outcomes.

7.2 Mechanism

The effective sample size is much smaller than the number of observations. Serially correlated residuals and persistent treatment indicators create repeated information that OLS treats as independent.

7.3 Evidence chain

The paper documents common risky practices in published DD work, demonstrates overrejection with placebo laws, confirms the mechanism in AR(1) simulations, and evaluates several corrections.

7.4 Practical implications

Researchers should cluster at the level of treatment variation when enough clusters exist, use aggregation or small-sample methods when clusters are few, and routinely examine residual autocorrelation and placebo tests.

7.5 What not to overinterpret

The paper does not show that DD point estimates are biased. It shows that naive standard errors are often invalid.

7.6 Final takeaway

The number of observations is not the number of independent pieces of identifying information. Inference must match the research design.